

■■ VAJRA AI THREAT INTELLIGENCE PLATFORM

Comprehensive Project Report: Autonomous Ransomware Defense & Enterprise Risk Monitoring

EXECUTIVE SUMMARY

VAJRA is an enterprise-grade AI Threat Intelligence and Risk Analysis platform engineered to combat escalating Ransomware-as-a-Service (RaaS) threats and perimeter vulnerabilities. By unifying 9 real-time threat feeds into a parallel processing pipeline, VAJRA calculates dynamic 0–100 Security Risk Scores, monitors live dark web ransomware victims, conducts 5-tab deep domain audits, and generates executive PDF and PowerPoint (.pptx) reports with a single click.

1. Project Introduction & Background

Cyber threats have evolved from isolated malware infections into systemic operational crises. Modern threat syndicates operate as **Ransomware-as-a-Service (RaaS)** businesses executing double-extortion tactics: exfiltrating proprietary enterprise data before encrypting core servers.

- **High-Frequency Targeting:** Over 1,200+ global cyber attacks occur daily.
- **Severe Financial Impact:** Average enterprise downtime cost exceeds \$1.85 Million per incident.
- **Perimeter Vulnerabilities:** Unpatched NVD CVEs, expired SSL/TLS certificates, and untrusted subdomains are heavily exploited.

2. Technical Architecture & Telemetry Pipeline

VAJRA utilizes a decoupled, high-performance architecture built with **FastAPI** (Python 3.11) on the backend and **Next.js 14** (React 18, TailwindCSS) on the frontend.

Layer	Technology / Service Component	Functionality
Ingestion	Asyncio, HTTPX Async Client	Parallel fetch across 9 external threat APIs
Threat Feeds	Ransomware.live, OTX, NIST NVD v2.0	Dark web victim radar & CVE vulnerability mapping
Core Backend	FastAPI, ReportLab, python-pptx	Risk score engine & multi-format report generation
Web Frontend	Next.js 14, TailwindCSS, TypeScript	Interactive 5-tab domain audit & ransomware radar

3. Deep Dive: Ransomware Intelligence Subsystem

The Ransomware Subsystem is VAJRA's premier defensive component, monitoring and profiling active ransomware syndicates in real time.

RaaS Group	Targeted Sectors	Telemetry Source
LockBit 3.0	Aviation, Logistics, Healthcare	Ransomware.live API v1/v2
BlackCat / ALPHV	Finance, Tech, Healthcare	Dark Web Victim Scraping
ClOp	Supply Chain & MSPs	NVD Zero-Day Vulnerability Map
Akira	Real Estate, Commercial Ops	AlienVault OTX Threat Pulses

4. Dynamic Risk Scoring & 5-Tab Domain Inspection

VAJRA features an automated **0–100 Security Risk Scoring Algorithm**. Starting at a baseline of 100 points, the score is penalized by expired SSL certificates (-30 pts), AbuseIPDB confidence > 50% (capped ≤50), URLScan malicious script detections (capped ≤45), and active NIST NVD CVE vulnerabilities (-15 pts per critical CVE).

- **Tab 1: Overview & IPs** - Resolved IPs, Geolocation, ISP owner, WHOIS domain age.
- **Tab 2: Domain Threats** - Identified threat categories, severity breakdown, and confidence ratings.
- **Tab 3: Vulnerabilities** - NIST NVD CVE mapping (``nvd.nist.gov/vuln/detail/CVE-...``) with CVSS v3 base scores.
- **Tab 4: SSL / TLS Audit** - Live socket handshake via Python ``certifi``, CA trust chain, and expiration countdown.
- **Tab 5: DNS Matrix** - A, MX, and TXT DNS records matrix.

5. Real-World Enterprise Use Cases

🚩 ■ Use Case 1: Aviation Security (e.g., IndiGo Airlines)

Commercial airlines process millions of daily flight bookings and payment API requests. VAJRA continuously audits the digital perimeter ('goindigo.in'), ensuring 100% SSL trust compliance, identifying open NVD CVEs on API gateways, and alerting SOC teams if partner vendors appear on ransomware leak sites.

■ Use Case 2: SOC Incident Response Acceleration

SOC analysts waste hours querying separate databases during incidents. VAJRA's 5-tab domain inspection and one-click PDF/PPTX export engine reduce investigation time by **90%**, delivering immediate C-Suite visibility.

6. Strategic Business Benefits for CISO & Leadership

1. **Proactive Risk Reduction:** Eliminates perimeter weaknesses before ransomware deployment.
2. **Consolidated Telemetry:** Merges 9 feeds into a single unified operational dashboard.
3. **Boardroom Visibility:** Standardized 0–100 Security Scores and instant executive PDF/PPTX downloads.
4. **Supply Chain Security:** Continuous third-party vendor domain auditing prevents vendor breaches.